

AE4L GATE

Dhruv Kakade

v26.0.0.1

Contents

1	Syllabus For Gate AE 2027	9
1.1	Maths	10
1.1.1	Linear Algebra	10
1.1.2	Calculus	10
1.1.3	Differential Equations	10
1.1.4	Numerical Methods	10
1.1.5	Special Topics	10
1.2	Aerodynamics	10
1.2.1	Basic fluid mechanics	10
1.2.2	Incompressible Flow	10
1.2.3	Compressible flows	11
1.2.4	Special Topics	11
1.3	Gas Turbines	11
1.3.1	Aerothermodynamics of aircraft engines	11
1.3.2	Engine performance	11
1.3.3	Aerothermodynamics of non-rotating propulsion components	11
1.3.4	Gas turbine combustor	11
1.3.5	Turbomachinery	11
1.3.6	Special Topics	12
1.4	Rocket Propulsion	12
1.5	Flight Mechanics	12
1.6	Orbital Mechanics	12
1.7	Strength Of Materials	12
1.8	Aircraft Structures	12
1.9	Mechanical Vibration and Aero-elasticity	12
1.10	Aptitude	12
2	Reference Books	13
2.1	Gas dynamics	14
2.2	Fluid mechanics	14
3	Unit Conversions and Constants	15
4	Maths	17
5	Aerodynamics	19
6	Gas Turbines	21

7	Rocket Propulsion	23
8	Flight Mechanics	25
9	Orbital Mechanics	27
10	Strength of Materials	29
11	Aircraft Structures	31
12	Mechanical Vibration and Aero-elasticity	33

List of Tables

List of Figures

Chapter 1

Syllabus For Gate AE 2027

1.1 Maths

1.1.1 Linear Algebra

Vector algebra, matrix algebra, systems of linear equations, rank of a matrix; Eigenvalues and eigenvectors.

1.1.2 Calculus

Functions of single variable, limits, continuity and differentiability, chain rule, maxima and minima, Integration; Functions of several variables, partial derivatives, gradient, divergence and curl, directional derivatives; Line, surface and volume integrals. Theorems of Stokes, Gauss and Green.

1.1.3 Differential Equations

First order linear ordinary differential equations; Higher order linear ODEs with constant coefficients; Classification of partial differential equations; *Solution to:* wave equation, Laplace equation, heat equation using separation of variables methods.

1.1.4 Numerical Methods

Numerical solutions for linear and nonlinear algebraic equations by bisection, Newton-Raphson method; basic numerical differentiation; numerical integration by trapezoidal and Simpson's rules; linear regression and least squares method; linear interpolation.

1.1.5 Special Topics

Fourier series; complex numbers, analytic functions and Cauchy-Riemann equations. Basics of probability and statistics: Bayes theorem, mean, median and mode; variance; binomial, normal and Poisson distribution.

1.2 Aerodynamics

1.2.1 Basic fluid mechanics

Fluid kinematics, streamline, streakline, pathline; conservation laws: mass, linear momentum and energy (integral and differential form); dimensional analysis and dynamic similarity; incompressibility conditions; Newtonian fluids.

1.2.2 Incompressible Flow

Elementary ideas of viscous flows

Hagen-Poiseuille flow, Couette flow, basic concepts in boundary layers, boundary layer thickness. Two-dimensional potential flow theory: sources, sinks, doublets, point vortex and their superposition, Bernoulli's equation.

Airfoils

Airfoil nomenclature; aerodynamic coefficients: lift, drag and moment; Kutta-Joukowski theorem; thin airfoil theory, Kutta condition, starting vortex;

Wings

Finite wing theory: induced drag, Prandtl lifting line theory; critical and drag divergence Mach numbers.

1.2.3 Compressible flows

Basic concepts of compressibility, one-dimensional compressible flows, isentropic flows, normal and oblique shocks, Prandtl-Meyer flow; flow through nozzles and diffusers.

1.2.4 Special Topics

Fanno flow; Rayleigh flow; pressure measurements using U-tube manometers and Pitot probe.

1.3 Gas Turbines

Basics of thermodynamics.

1.3.1 Aerothermodynamics of aircraft engines

Thrust, efficiency, range. Brayton cycle.

1.3.2 Engine performance

Ramjet, turbojet, turbofan, turboprop and turboshaft engines; after-burners.

1.3.3 Aerothermodynamics of non-rotating propulsion components

Intakes, nozzles.

1.3.4 Gas turbine combustor

Types and configurations; combustion, stoichiometric fuel-to-air ratio.

1.3.5 Turbomachinery**Axial compressors**

Angular momentum, work and compression, characteristic performance of a single axial compressor stage, efficiency of the compressor and degree of reaction, multi-staging;

Centrifugal compressor

Stage dynamics, inducer, impeller and diffuser; Axial turbines: stage performance.

1.3.6 Special Topics

Turbine blade cooling, compressor-turbine matching.

1.4 Rocket Propulsion

Rockets thrust equation and specific impulse, rocket performance; multi-staging; chemical rockets; performance of solid and liquid propellant rockets.

1.5 Flight Mechanics**1.6 Orbital Mechanics****1.7 Strength Of Materials****1.8 Aircraft Structures****1.9 Mechanical Vibration and Aero-elasticity****1.10 Aptitude**

Chapter 2

Reference Books

2.1 Gas dynamics

1. Fundamentals of Gas Dynamics by Robert D. Zucker, Oscar Biblarz
2. Fundamentals of Aerodynamics by John Anderson
3. Modern Compressible Flow: With Historical Perspective by John Anderson
4. Fundamentals Of Compressible Flow by SM Yahya
5. Aerodynamics for Engineering students by E. L. Houghton, N. B. Carruthers
6. Gas Dynamics by Ethirajan Rathakrishnan

2.2 Fluid mechanics

1. Fluid Mechanics: Fundamentals and Applications by Yunus A. Cengel, John M. Cimbala
2. Fluid Mechanics by Frank M. White
3. Fluid Mechanics by RK Bansal (for basics and more solved problems)

- Strength of materials - Strength of Materials by R.K. Rajput - Mechanics of Materials by James M. Gere - Strength Of Materials Parts I and II by Timoshenko S. - Strength of Materials by S. Ramamrutham

- Vibrations - Mechanical Vibrations by S.S Rao - Dynamics of Structures by Jagmohan L. Humar - Mechanical Vibrations by V.P Singh

- Structures - Aircraft Structures for Engineering Students By T.H.G. Megson - Analysis and Design of Flight Vehicle Structures by E. F Bruhn - Mechanics of Aircraft Structures by C. T. Sun - Analysis of Aircraft Structures: An Introduction by Bruce k. Donaldson

- Aircraft Propulsion - Gas Turbine Theory by Cohen, H., Rogers - Gas Turbines by V. Ganesan - Aircraft Propulsion and Gas Turbine Engines by Ahmed F. El-Sayed - Aircraft Propulsion by Saeed Farokhi

- Rocket Propulsion - Rocket Propulsion Elements by George P. Sutton - Propellants and Explosives by Naminosuke Kubota - Rocket Propulsion by K Ramamurthi

- Heat transfer basics - Heat and Mass Transfer by R.K. Rajput - Heat and Mass Transfer by JP holman

- Engineering Mathematics - Advanced Engineering Mathematics by Erwin Kreyszig - Advanced Engineering Mathematics by H. K. Dass - Higher Engineering Mathematics by B. S. Grewal

- Flight Mechanics - Aircraft Performance and Design by John Anderson - Flight Stability and Automatic Control by R. C. Nelson - Airplane Performance, Stability and Control by Courtland D. Perkins, Robert E. Hage - Introduction to Flight by John Anderson - Aerodynamics by Clancy

- Space Mechanics - Orbital Mechanics for Engineering Students by Howard D. Curtis - Fundamentals of Astrodynamics by Donald D. Mueller, Jerry White, Roger R. Bate - Spaceflight Dynamics by William E. Wiesel - Orbital Mechanics by John E. Prussing, Bruce A. Conway - Analytical Mechanics of Space Systems by Hanspeter Schaub, John L. Junkins

- Previous Year Questions - GK Publisher

¿ MIT OCW / NPTEL - Assignments and Problem Sets ¿ The main books in focus are discussed below (soon). All the books mentioned need not be used.

Chapter 3

Unit Conversions and Constants

Chapter 4

Maths

Chapter 5

Aerodynamics

Chapter 6

Gas Turbines

Chapter 7

Rocket Propulsion

Chapter 8

Flight Mechanics

Chapter 9

Orbital Mechanics

Chapter 10

Strength of Materials

Chapter 11

Aircraft Structures

Chapter 12

Mechanical Vibration and Aero-elasticity